

Name: _____

Answer all questions covering the introduction to file systems.

1. Suppose I have a directory tree of a number of files and directories. Answer the following questions about this directory tree.

- (a) Can directories be leaves of the directory tree? Explain.

- (b) Which are represented with an inode number : files, directories, or both?

2. Suppose I write to a `tmp` file for checkpointing (backing up the state of my application as it is running). At the end of my program, I truncate the file to 0, deleting the current contents. After my program ends, I see the `tmp` file has data remaining in it. Explain how this could happen.

3. Why is renaming a file an atomic operation?